

Recommendation Memo: A/B Test Decision for Closed-Domain RAG System

Executive Summary

Based on the A/B test simulation, Variant B is recommended for deployment. It delivers statistically significant improvements in correctness and groundedness, and reduces hallucination rates, despite a moderate increase in latency.

Key Findings

Correctness improved from 79.2% to 88.0% ($p = 0.00788$).

Groundedness improved from 83.2% to 92.0% ($p = 0.00283$).

Hallucination rate decreased from 8.4% to 3.6%.

Latency increased from 2.10s to 2.58s on average.

Interpretation

Variant B provides a clear improvement in answer quality and reliability. The reduction in hallucination risk and increase in groundedness significantly enhance trust in system outputs. While latency increases, it remains within a reasonable range for this application.

Recommendation

Deploy Variant B. The quality improvements outweigh the latency cost. Monitor latency and optimize performance post-deployment if necessary.